Canonical Bundle Formula

We work over an algebraically closed field of characteristic zero, like $\mathbb{C}$. All varieties are assumed to be normal unless otherwise stated.

1 Birational divisors

Definition 1 (b-divisors)**.** Let X be a normal algebraic variety. We call Y a *birational model over* X, or simply a *model over* X, if there exists a proper birational morphism $Y \to X$ of normal varieties. When both X and Y are quasi-projective, $Y \to X$ is projective.

Let $X' \dashrightarrow X$ be a birational map. For any prime divisor E (resp. algebraic valuation v) over X, we denote by $c_{X'}(E)$ (resp. $v_{X'}$) the centre of E (resp. v) on X', whenever it exists. A *b-divisor* $\mathbf{D}$ on X is a formal sum

$$\mathbf{D} = \sum_E r_E E \qquad \left(\text{resp. } \sum_v r_v v\right),$$

where the E are prime divisors over X (resp. the v are algebraic valuations over X) and $r_E \in \mathbb{R}$ (resp. $r_v \in \mathbb{R}$), such that only finitely many terms with nonzero coefficient have a divisorial centre on X. If, in addition, $r_E \in \mathbb{Q}$ for every E (resp. $r_v \in \mathbb{Q}$ for every v), then $\mathbf{D}$ is called a $\mathbb{Q}$-*b-divisor* on X.

The *trace* of $\mathbf{D}$ on a model $X' \to X$ is the $\mathbb{R}$-divisor

$$\operatorname{tr}_{X'} \mathbf{D} = D_{X'} = \sum_{E' \text{ a divisor on } X'} r_{E'} E' \qquad \left(\text{resp. } \sum_{v_{X'} \text{ a divisor}} r_v v_{X'}\right).$$

If $D_{X'}$ is $\mathbb{R}$-Cartier and D_Y is the pullback of $D_{X'}$ on Y for every model $Y \to X'$, we say that $\mathbf{D}$ *descends to* X' and write

$$\mathbf{D} = \overline{D_{X'}}.$$

Let $X \to U$ be a proper morphism, and assume that $\mathbf{D}$ is a b-divisor over X which descends to some model $Y \to X$. If D_Y is nef$/U$ (resp. semi-ample$/U$), then we say that $\mathbf{D}$ is *b-nef*$/U$ (resp. *b-semi-ample*$/U$). If D_Y is an $\mathbb{R}$-Cartier divisor, then we say that $\mathbf{D}$ is $\mathbb{R}$-*b-Cartier*. The terms b-Cartier and $\mathbb{Q}$-b-Cartier are defined similarly using Cartier and $\mathbb{Q}$-Cartier divisors, respectively. If $\mathbf{D}$ can be written as an $\mathbb{R}_{\geq 0}$-linear combination of nef$/U$ $\mathbb{Q}$-b-Cartier b-divisors, then we say that $\mathbf{D}$ is NQC/U.

When $U = \operatorname{Spec} k$, this gives the absolute setting.

Definition 2 (Generalized pairs)**.** A *generalized sub-pair* (or *g-sub-pair* for short) is a triple $(X, B, \mathbf{M})/U$ consisting of a normal algebraic variety X associated with a proper morphism $X \to U$, an $\mathbb{R}$-divisor B on X, and an $\mathbb{R}$-b-Cartier b-nef$/U$ divisor $\mathbf{M}$ on X, such that $K_X + B + M_X$ is $\mathbb{R}$-Cartier. Note that, on a model $h : W \to X$ on which $\mathbf{M}$ descends, we put

$$M_X := h_* M_W.$$

If B is a $\mathbb{Q}$-divisor and $\mathbf{M}$ is $\mathbb{Q}$-b-Cartier, then we say that $(X, B, \mathbf{M})/U$ is a $\mathbb{Q}$-*g-sub-pair*.

If $\mathbf{M} = 0 = \overline{0}$, a g-sub-pair $(X, B, \mathbf{M})/U$ is called a *sub-pair*. If $U = \{\text{pt}\}$, or if U is not important, we drop U from the notation.

A g-sub-pair (resp. $\mathbb{Q}$-g-sub-pair) $(X, B, \mathbf{M})/U$ is called a *g-pair* (resp. $\mathbb{Q}$-*g-pair*) if $B \geq 0$. A sub-pair (X, B) is called a *pair* if $B \geq 0$.

Definition 3 (Singularities of g-pairs)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair. For any prime divisor E on X and any $\mathbb{R}$-divisor D on X, we define $\operatorname{mult}_E D$ to be the multiplicity of D along E.

Let $h : W \to X$ be any log resolution of (X, B) such that $\mathbf{M}$ descends to W, and write

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

The *log discrepancy* of a prime divisor E on W with respect to $(X, B, \mathbf{M})$ is $1 - \operatorname{mult}_E B_W$, and it is denoted by $A(E, X, B, \mathbf{M})$. For the discrepancy

$$a(E, X, B, \mathbf{M}) = -\operatorname{mult}_E B_W,$$

we have

$$A(E, X, B, \mathbf{M}) = 1 + a(E, X, B, \mathbf{M}).$$

We say that $(X, B, \mathbf{M})$ is *sub-g-lc* (resp. *sub-g-klt*) if

$$A(E, X, B, \mathbf{M}) \geq 0 \qquad (\text{resp. } > 0)$$

for every log resolution $h : W \to X$ as above and every prime divisor E on W.

Suppose that $(X, B, \mathbf{M})$ is sub-g-lc. A *g-lc place* of $(X, B, \mathbf{M})$ is a prime divisor E over X such that

$$A(E, X, B, \mathbf{M}) = 0.$$

A *g-lc centre* of $(X, B, \mathbf{M})$ is the centre of a g-lc place on X. We often say *lc place* and *lc centre* for short when the pair is clear.

We say that $(X, B, \mathbf{M})$ is *(sub-)g-dlt* if it is (sub-)g-lc and there exists an open subset $U \subseteq X$ such that

(1) $(U, B|_U)$ is log smooth;

(2) for any lc place E over X, we have $c_X(E) \cap U \neq \varnothing$, and $c_X(E) \cap U$ is an lc centre of $(U, B|_U)$.

There is an important result in [Bir-Zhang14, proof of Lemma 4.5], also proved in [HL22, Proposition 3.9].

Definition–Lemma 4 (G-dlt modifications)**.** Let $(X, B, \mathbf{M})/U$ be a quasi-projective g-lc g-pair. Then there is a projective birational morphism $r : X' \to X$ and a g-lc g-pair $(X', B', \mathbf{M})/U$ such that

(1) $(X', B', \mathbf{M})$ is $\mathbb{Q}$-factorial g-dlt;

(2) $K_{X'} + B' + M_{X'} = r^*(K_X + B + M_X)$; and

(3) any r-exceptional divisor is a component of $\lfloor B' \rfloor$.

For any projective birational morphism r and $(X', B', \mathbf{M})/U$ which satisfy (1)–(3), $(r, (X', B', \mathbf{M})/U)$ will be called a *g-dlt modification* of $(X, B, \mathbf{M})$.

It has a useful application; see [JLX22, Lemma 2.6] or my note [Adj, 9, 10, 14, 15].

Definition–Lemma 5 (G-dlt adjunction)**.** Let $(X, B, \mathbf{M})/U$ be a g-lc g-pair and let $S \subseteq \lfloor B \rfloor$ be a prime divisor, with normalization $\nu : S^\nu \to S$. Then there is a naturally defined g-pair structure $(S^\nu, B_{S^\nu}, \mathbf{M}^S)/U$ such that

$$K_{S^\nu} + B_{S^\nu} + M^S_{S^\nu} \sim_{\mathbb{R}} (K_X + B + M_X)|_{S^\nu}.$$

We say that $(S^\nu, B_{S^\nu}, \mathbf{M}^S)/U$ is the g-pair obtained by *divisorial adjunction*. If $(X, B, \mathbf{M})$ is g-lc, then $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is g-lc, and if $(X, B, \mathbf{M})$ is g-dlt, then $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is g-dlt.

For the construction of divisorial adjunction, see [Adj, 9 and 10]; for the g-dlt and g-lc parts, see [Adj, 14 and 15].

Definition 6 (Discrepancy b-divisors)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair. For each model $f : Y \to X$, choose compatible canonical divisors with $f_*K_Y = K_X$, and set

$$\operatorname{tr}_Y \mathbf{A}(X, B, \mathbf{M}) = K_Y + M_Y - f^*(K_X + B + M_X).$$

These traces define the *discrepancy b-divisor* $\mathbf{A}(X, B, \mathbf{M})$; we often write its trace A_Y for short. Define another b-divisor by

$$\operatorname{tr}_Y \mathbf{A}^*(X, B, \mathbf{M}) = A_Y^{>-1},$$

where we omit the components with coefficient at most -1. For a prime divisor E over X,

$$\operatorname{mult}_E \mathbf{A}(X, B, \mathbf{M}) = A(E, X, B, \mathbf{M}) - 1 = a(E, X, B, \mathbf{M}),$$

where $a(E, X, B, \mathbf{M})$ is the discrepancy of E.

Definition 7 (Sheaf associated to a b-divisor)**.** Let $\mathbf{D}$ be an $\mathbb{R}$-b-divisor on X. The sheaf $\mathcal{O}_X(\mathbf{D})$ is an $\mathcal{O}_X$-module whose sections on an open subset $U \subseteq X$ are given by

$$H^0\big(U, \mathcal{O}_X(\mathbf{D})\big) = \big\{\varphi \in k(X)^* \;\big|\; \operatorname{div}_Y(\varphi) + D_Y \geq 0 \text{ on } h^{-1}(U) \text{ for every model } h : Y \to X\big\} \cup \{0\},$$

where $k(X)$ is the function field of X. Note that $\mathcal{O}_X(\mathbf{D})$ is not necessarily coherent.

Lemma 8. *When* $\mathbf{D}$ *is b-Cartier,* $\mathcal{O}_X(\mathbf{D})$ *is coherent.*

Proof. Take a model $h : W \to X$ on which $\mathbf{D}$ descends. Then we have

$$\mathcal{O}_X(\mathbf{D}) = h_*\mathcal{O}_W(D_W).$$

Indeed, the inclusion $\mathcal{O}_X(\mathbf{D}) \subseteq h_*\mathcal{O}_W(D_W)$ is trivial. For the other inclusion, suppose that $\varphi \in H^0(h^{-1}(V), \mathcal{O}_W(D_W))$. This means that

$$\operatorname{div}_W(\varphi) + D_W \geq 0 \qquad \text{on } h^{-1}(V).$$

For any other model $Y \to X$, take a high common model

$$W \xleftarrow{\ r\ } T \xrightarrow{\ s\ } Y$$

over X. Since $\mathbf{D}$ descends to W,

$$\operatorname{div}_T(\varphi) + D_T = r^*\big(\operatorname{div}_W(\varphi) + D_W\big) \geq 0$$

on $(h \circ r)^{-1}(V)$. Pushing forward via s gives

$$\operatorname{div}_Y(\varphi) + D_Y = s_*\big(\operatorname{div}_T(\varphi) + D_T\big) \geq 0$$

on $s((h \circ r)^{-1}(V))$. Thus $h_*\mathcal{O}_W(D_W) \subseteq \mathcal{O}_X(\mathbf{D})$, and coherence follows from the properness of h. □

For a b-divisor $\mathbf{D}$ on X, its *round up* $\lceil \mathbf{D} \rceil$ is defined by

$$\mathrm{tr}_Y \lceil \mathbf{D} \rceil = \lceil D_Y \rceil$$

on every model $Y \to X$. Similarly for round down.

Definition 9 (Contraction)**.** A *contraction* is a proper surjective morphism $f : X \to Y$ of normal varieties with $f_* \mathcal{O}_X = \mathcal{O}_Y$.

Equivalently, f has connected fibres. Indeed, in the Stein factorization

$$X \longrightarrow \widetilde{Y} \longrightarrow Y,$$

connected fibres force the finite morphism $\widetilde{Y} \to Y$ to have degree one, i.e., it is a birational morphism between normal varieties, so it is an isomorphism.

Definition 10 (G-lc-trivial fibration)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair, and let $f : X \to Y$ be a contraction over U. We call $f : (X, B, \mathbf{M}) \to Y$ a *g-lc-trivial fibration* over U if

(1) $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of Y;

(2) $\operatorname{rank} f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = 1$; and

(3) there exists an $\mathbb{R}$-Cartier divisor L on Y such that

$$K_X + B + M_X \sim_{\mathbb{R}} f^* L.$$

If, in addition, $(X, B, \mathbf{M})$ is g-sub-klt over the generic point of Y, we say *g-klt-trivial*. For $\mathbf{M} = 0$, these are classical lc-trivial and klt-trivial fibrations; see [FG14, Definitions 3.1–3.2].

Remark 11. Condition (3) is sometimes written as

$$K_X + B + M_X \sim_{\mathbb{R}, Y} 0.$$

When f is a contraction, they are equivalent.

Lemma 12 ([Fuj12, Lemmas 3.19–3.20])**.** *Let $(X, B, \mathbf{M})/U$ be a g-sub-lc g-sub-pair, and let $h : W \to X$ be a log resolution on which $\mathbf{M}$ descends. Write*

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

Then

$$\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big).$$

In particular, this sheaf is coherent. If $B \geq 0$, then

$$\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = \mathcal{O}_X.$$

Proof. By definition,

$$\mathrm{tr}_W \, \mathbf{A}(X, B, \mathbf{M}) = -B_W, \qquad \mathrm{tr}_W \, \mathbf{A}^*(X, B, \mathbf{M}) = -B_W^{<1}.$$

Let $g : W' \to W$ be any model over W. Define $B_{W'}$ by

$$K_{W'} + B_{W'} + M_{W'} = (h \circ g)^*(K_X + B + M_X).$$

Since $\mathbf{M}$ descends to W, we have $M_{W'} = g^* M_W$. Cancelling these terms gives

$$K_{W'} + B_{W'} = g^*(K_W + B_W).$$

Step 1. We claim that

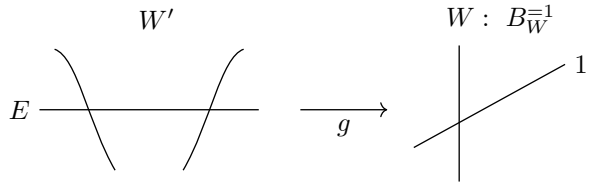
$$\lceil \mathbf{A}^*(X, B, \mathbf{M})_{W'} \rceil = g^* \lceil \mathbf{A}^*(X, B, \mathbf{M})_W \rceil + F,$$

where $F \geq 0$ is an integral g-exceptional divisor.

The divisor $\lceil -B_W^{<1} \rceil$ is an effective Cartier divisor, and is supported on the negative components of B_W. Thus the pair

$$\big(W, B_W + \lceil -B_W^{<1} \rceil\big)$$

is log smooth and has lc singularities, since for a coefficient $b < 1$, we have $0 \leq b + \lceil -b \rceil < 1$. It also has the same lc places as (W, B_W), because an lc place of either log smooth pair has centre a stratum of $B_W^{=1}$, but $\operatorname{Supp} \lceil -B_W^{<1} \rceil$ contains no such stratum.



$$a(E) = 2 - 1 - (1 + 1) = -1$$

For a prime divisor E on W', the number $\operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil$ is a nonnegative integer, since $g^* \lceil -B_W^{<1} \rceil \geq 0$. If $\operatorname{mult}_E B_{W'} = 1$, then E is an lc place of (W, B_W), so its multiplicity in $g^* \lceil -B_W^{<1} \rceil$ is zero. Both sides of the claimed equality

$$\lceil -B_{W'}^{<1} \rceil = g^* \lceil -B_W^{<1} \rceil + F$$

then give $\operatorname{mult}_E F = 0$.

If $\operatorname{mult}_E B_{W'} < 1$, we have

$$A(E, W, B_W) = 1 - \operatorname{mult}_E B_{W'} > 0,$$

so E is not an lc place of (W, B_W). Since the two pairs have the same lc places, we have

$$\begin{aligned} 0 &< A\big(E, W, B_W + \lceil -B_W^{<1} \rceil\big) \\ &= 1 - \operatorname{mult}_E B_{W'} - \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil. \end{aligned}$$

Rounding this inequality gives

$$\lceil -\operatorname{mult}_E B_{W'} \rceil \geq \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil \geq 0.$$

The left-hand side is nothing but $\operatorname{mult}_E \lceil -B_{W'}^{<1} \rceil$, so $\operatorname{mult}_E F \geq 0$. Since (W, B_W) is sub-lc, the coefficients of $B_{W'}$ are at most 1. Therefore $F \geq 0$ is effective. Since both sides of the claimed equality are integral divisors, F is integral.

Finally, if E is not g-exceptional, then $g(E)$ is a divisor on W. That is, g is an isomorphism over the generic point of $g(E)$ by normality, so we have

$$\operatorname{mult}_E B_{W'} = \operatorname{mult}_{g(E)} B_W.$$

Then

$$\operatorname{mult}_E F = \operatorname{mult}_E \lceil -B_{W'}^{<1} \rceil - \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil = 0.$$

We see that F is g-exceptional. This proves the claim.

Step 2. By algebraic Hartogs, we have

$$g_* \mathcal{O}_{W'}(F) = \mathcal{O}_W.$$

Then we have

$$\begin{aligned} g_* \mathcal{O}_{W'}\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})_{W'} \rceil\big) &= g_* \mathcal{O}_{W'}\big(g^* \lceil -B_W^{<1} \rceil + F\big) \\ &\cong \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big), \end{aligned}$$

thanks to the projection formula. These identifications are induced by the identity on rational functions.

For any other model $\psi : Y \to X$, take a common higher model

$$W \xleftarrow{r} T \xrightarrow{s} Y.$$

Fix any open subset $V \subseteq X$. Suppose that we have

$$\operatorname{div}_W(\varphi) + \lceil -B_W^{<1} \rceil \geq 0$$

on $h^{-1}(V)$. Applying the claimed equality to $r : T \to W$, we have

$$r_* \mathcal{O}_T\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil\big) \cong \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big).$$

Thus it gives

$$\operatorname{div}_T(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil \geq 0$$

on $(h \circ r)^{-1}(V)$. We push forward this along $s : T \to Y$ and get

$$\operatorname{div}_Y(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_Y \rceil = s_*\big(\operatorname{div}_T(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil\big) \geq 0$$

on $s((h \circ r)^{-1}(V)) = \psi^{-1}(V)$. It gives

$$h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big) \subseteq \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big).$$

The reverse direction is trivial. Coherence is due to the properness of h.

Finally, if $B \geq 0$, then every non-exceptional component of B_W has coefficient in $[0, 1]$. Hence $\lceil -B_W^{<1} \rceil$ is effective and h-exceptional. By algebraic Hartogs, we have

$$h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big) = \mathcal{O}_X.$$

□

Now let $f : X \to Y$ be a contraction over U. If $(X, B, \mathbf{M})$ is only g-sub-lc over the generic point of Y, i.e., it is g-sub-lc over an open subset $Y^0 \subseteq Y$, applying Lemma 12, we have

$$\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = h_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big)$$

on $f^{-1}(Y^0)$. Pushing forward along f, we get

$$f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = (f \circ h)_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big)$$

on Y^0, and since $\eta_Y \in Y^0$, we have

$$\operatorname{rank} f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = \operatorname{rank}(f \circ h)_* \mathcal{O}_W\big(\lceil -B_W^{<1} \rceil\big).$$

Lemma 13 (The rank-one condition)**.** *Let $(X, B, \mathbf{M})/U$ be a g-sub-pair, and let $f : X \to Y$ be a contraction over U. Suppose that $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of Y and that B is effective over that point, i.e., its components which dominate Y have nonnegative coefficients. Then*

$$\operatorname{rank} f_*\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})\rceil\big) = 1.$$

Proof. We may replace Y by an open subset and assume that $B \geq 0$ and $(X, B, \mathbf{M})$ is g-lc; this does not change the rank. Take $h : W \to X$ to be a log resolution on which $\mathbf{M}$ descends. The divisor $\lceil -B_W^{<1}\rceil$ is effective and h-exceptional. So by Lemma 12,

$$\operatorname{rank} f_*\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})\rceil\big) = \operatorname{rank} f_*\mathcal{O}_X = \operatorname{rank} \mathcal{O}_Y = 1.$$

□

Example 14 ($B \not\geq 0$ over the generic point)**.** Put $f : \mathbb{P}^1 \to \operatorname{Spec} \mathbb{C}$, $B = -p$, and $\mathbf{M} = 0$, where $p \in \mathbb{P}^1$. The pair $(\mathbb{P}^1, B)$ is a sub-klt pair. Take $h = \mathrm{id}$. Then

$$h_*\mathcal{O}_{\mathbb{P}^1}\big(\lceil -B^{<1}\rceil\big) = \mathcal{O}_{\mathbb{P}^1}(p).$$

Therefore,

$$\operatorname{rank} f_*\mathcal{O}_{\mathbb{P}^1}(p) = \dim_{\mathbb{C}} H^0\big(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(p)\big) = 2,$$

not rank one.

2 Divisorial Thresholds: Discriminant and Moduli b-divisor

For a contraction $f : (X, B, \mathbf{M}) \to Y$ where $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of Y, we fix an $\mathbb{R}$-Cartier divisor L on Y and an $\mathbb{R}$-linear equivalence

$$K_X + B + M_X \sim_{\mathbb{R}} f^*L.$$

Let P be a prime divisor over Y. Choose a smooth model $\psi : Y' \to Y$ on which P appears. Resolve the main component of $X \times_Y Y'$ and take a sufficiently high model W on which $\mathbf{M}$ descends. We obtain the commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{h} & X \\ {\scriptstyle f_W}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[\psi]{} & Y. \end{array}$$

We put

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

Definition 15. The *generalized lc threshold* over η_P, the generic point of P, is

$$t_P = \sup\big\{s \in \mathbb{R} \mid (W, B_W + sf_W^*P, \mathbf{M}) \text{ is g-sub-lc over } \eta_P\big\}.$$

The condition is tested over an open neighbourhood of η_P in Y', and the parameter s may be negative.

Lemma 16 (Threshold formula)**.** *After replacing W by a higher log resolution, we may assume that* $\operatorname{Supp} B_W \cup \operatorname{Supp} f_W^* P$ *is SNC over η_P. Let $E_1, \dots, E_n$ be the components of this union meeting the fibre $f_W^{-1}(\eta_P)$, and put*

$$b_i = \operatorname{mult}_{E_i} B_W, \qquad m_i = \operatorname{mult}_{E_i} f_W^* P.$$

Then

$$t_P = \min_{m_i > 0} \frac{1 - b_i}{m_i}. \tag{2.1}$$

The value of t_P is independent of the choice of log resolution.

Proof. Since $\mathbf{M}$ descends to W, the generalized discrepancies coincide with the ordinary discrepancies of $(W, B_W + s f_W^* P)$. The SNC criterion gives that this sub-pair is sub-lc over η_P if and only if $b_i + s m_i \le 1$ for every i. Since f_W is proper and E_i meets the fibre over η_P, its image is either P or Y'. If $m_i = 0$, the component E_i dominates Y', and $b_i \le 1$ by generic sub-lc. If $m_i > 0$, the condition is $s \le (1 - b_i)/m_i$. At least one such component exists because f_W is surjective and P is an effective Cartier divisor. This proves (2.1).

On a common higher resolution over Y', the crepant pullbacks of $K_W + B_W + s f_W^* P + M_W$ compute the same discrepancies. Hence t_P is independent of the resolution. See [BGR, Lemma 5.2.16] for the crepant pullback case of the ordinary discrepancy comparison. □

Definition 17 (Discriminant and moduli part)**.** For each prime divisor P over Y, put $\operatorname{mult}_P \mathbf{C} = 1 - t_P$. On a smooth model $\psi : Y' \to Y$, define

$$C_{Y'} = \sum_{P \subset Y'} (1 - t_P) P, \qquad N_{Y'} = \psi^* L - K_{Y'} - C_{Y'},$$

where the sum runs over the prime divisors on Y'. The b-divisor $\mathbf{C}$ is called the *discriminant*, and $\mathbf{N} = \overline{L} - \mathbf{K} - \mathbf{C}$ is the *moduli b-divisor*, where $\mathbf{K}$ has traces given by compatible canonical divisors on the models of Y.

On other models their traces are obtained by pushforward from a dominating smooth model.

Proposition 18. $\mathbf{C}$ *and* $\mathbf{N}$ *are b-divisors, and on every model $\psi : Y' \to Y$ their traces satisfy*

$$K_{Y'} + C_{Y'} + N_{Y'} = \psi^* L.$$

Proof. Step 1. Fix a smooth model $\psi : Y' \to Y$. Choose a log resolution $h : W \to X$ mapping to Y' on which $\mathbf{M}$ descends and $\operatorname{Supp} B_W$ is SNC. We use the same diagram

$$\begin{array}{ccc} W & \xrightarrow{h} & X \\ {\scriptstyle f_W}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[\psi]{} & Y. \end{array}$$

By generic smoothness, there is a nonempty open subset $U \subseteq Y'$ such that $f_W^{-1}(U) \to U$ is smooth. By properness of f_W, we may shrink U to cut off the images of the vertical components of B_W. Applying generic smoothness to the strata of the remaining horizontal support and removing the images of the strata which do not dominate Y', we may also

assume that all their nonempty intersections are smooth over U. Thus the horizontal support is relative SNC over U.

For a prime divisor P on Y' meeting U, its generic point η_P lies in U. Take an open subset $\eta_P \in V \subseteq U$ such that $P_V = P \cap V$ is smooth, and set $W_V = f_W^{-1}(V)$. The pullback divisor over V has the following scheme-theoretic description.[1]

$$D := f_W^* P|_{W_V} \cong (P \times_{Y'} W) \times_W (V \times_{Y'} W) \cong W_V \times_V P_V.$$

Since $W_V \to V$ is smooth, $D \to P_V$ is smooth. As P_V is smooth, D is smooth and hence reduced. In particular, $f_W^* P$ is reduced over η_P.

At any point of D, let $H_1, \dots, H_r$ be the components of $\operatorname{Supp} B_W$ passing through that point, and set $S = \bigcap_{i=1}^r H_i$ (with $S = W$ if $r = 0$). Write $S_V = S \cap W_V$. We have

$$S \cap D \cong S \times_W (W_V \times_V P_V) \cong S \times_{Y'} P_V \cong S_V \times_V P_V.$$

This is the scheme-theoretic inverse image of P_V under $f_W|_{S_V} : S_V \to V$. Since $S \cap f_W^{-1}(U) \to U$ is smooth by the relative SNC condition, $S_V \to V$ is smooth. Thus $S \cap D$ is smooth and, where nonempty, is an effective Cartier divisor in S_V, by smooth base change of the smooth Cartier divisor P_V. Therefore D meets the horizontal boundary strata transversely.

On W_V write

$$B_W|_{W_V} = \sum_i b_i H_i, \qquad D = \sum_j D_j,$$

where D is reduced and the H_i are the horizontal components restricted to W_V. Each D_j maps into P_V, so no D_j is a component of $B_W|_{W_V}$. Then

$$B_W|_{W_V} + sD = \sum_i b_i H_i + \sum_j s D_j,$$

where every $b_i \le 1$ because $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of Y. Since $\mathbf{M}$ descends to W, the g-sub-lc condition over η_P is equivalent to the ordinary sub-lc condition for the boundary on the right. This boundary has SNC support, and every D_j dominates P_V by smoothness and properness. There is at least one such component. Thus the condition is equivalent to $s \le 1$, giving $t_P = 1$. Only the finitely many prime divisors in $Y' \setminus U$ can contribute to $C_{Y'}$.

Step 2. Let $r : Y'' \to Y'$ be a higher smooth model. If $\widetilde{P}$ is the strict transform of P, they define the same divisorial valuation. Since r is an isomorphism over a neighbourhood of η_P, the thresholds agree: $t_{\widetilde{P}} = t_P$. The r-exceptional terms push forward to zero, so

$$r_* C_{Y''} = \sum_{P \subset Y'} (1 - t_{\widetilde{P}}) r_* \widetilde{P} = \sum_{P \subset Y'} (1 - t_P) P = C_{Y'}.$$

Choose compatible canonical divisors, i.e., traces of $\mathbf{K}$. Then

$$r_* N_{Y''} = r_* \big(r^* \psi^* L - K_{Y''} - C_{Y''} \big) = \psi^* L - K_{Y'} - C_{Y'} = N_{Y'}.$$

A common higher smooth model therefore makes the pushforwards to any normal model independent of the smooth model used to compute them. This proves that $\mathbf{C}$ and $\mathbf{N}$ are b-divisors. Pushing forward the identity on a smooth model also gives the stated identity on every model. $\square$

[1] Here we use the scheme-theoretic notation $P_V = P \cap V \cong P \times_{Y'} V$ and $W_V = f_W^{-1}(V) \cong V \times_{Y'} W$. The displayed identification follows by transitivity of fibre products [Görtz I, Proposition 4.16].

Note that if

$$L' = L + \sum_j r_j \operatorname{div}_Y(u_j), \qquad r_j \in \mathbb{R}, \quad u_j \in k(Y)^\times,$$

then the same discriminant $\mathbf{C}$ occurs for L' and

$$N'_{Y'} = N_{Y'} + \sum_j r_j \operatorname{div}_{Y'}(u_j)$$

on each model $\psi : Y' \to Y$, where the u_j are viewed as rational functions on Y'. Thus changing L by an $\mathbb{R}$-principal divisor changes $\mathbf{N}$ by the corresponding $\mathbb{R}$-principal b-divisor.

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